# Mojtaba Hajian

Bolzano, Italy | mojtabambb@gmail.com | LinkedIn | Google Scholar

## Profile

**AI and Machine Learning Researcher** with a Ph.D. in ICT, focusing on scalable machine learning, representation learning, and data-driven modeling over high-dimensional and structured data. Experienced in developing and evaluating supervised, unsupervised, and deep learning algorithms across multiple domains, including medical imaging, biomedical signals, and large-scale analytical systems. Strong background in ML research and reproducible pipelines, with publications in peer-reviewed venues.

## Experience

- **RESEARCH ENGINEER (TECHNOLOGIST)| FREE UNIVERSITY OF BOZEN-BOLZANO, FACULTY OF ENGINEERING, ITALY | JULY 2026 – PRESENT**
- Research Technologist at unibz Cybersecurity Lab (CSLab)
- Supporting the deployment, customization, and security assessment of Cyber Range and MISP (Malware Information Sharing Platform and Threat Sharing) infrastructures
- Work spans offensive/defensive attack simulation and the development of AI-based models for network flow anomaly and intrusion detection.

- **RESEARCH FELLOW| FREE UNIVERSITY OF BOZEN-BOLZANO, FACULTY OF ENGINEERING, ITALY | JUNE 2025 – MAY 2026**
  - Working on the AIRFRAME project on AI-driven radiomics for metastatic castration-resistant prostate cancer (mCRPC), analysing *multi-modal PET/CT images* to predict therapy outcome and survival
  - Design and evaluate *machine-learning and deep-learning models*
  - Build end-to-end pipelines in *Python (PyRadiomics, SimpleITK)* for feature extraction, dimensionality reduction of handcrafted and radiomic features
  - Implement explainable-AI techniques to visualize feature importances and patient-specific outcome probabilities for clinicians
  - Writing professional reports and publishing in peer-review journals

- **PH.D. CANDIDATE| UNIVERSITY OF CALABRIA, DIMES, ITALY | JANUARY 2022 – MARCH 2025**
  - Leverage unsupervised learning methods to discover patterns within complex, high-dimensional data
  - Develop large-scale data analytics workflows utilizing Apache Spark, Hive, and Hadoop to support high-volume, low-latency data processing
  - Author scientific publications, technical reports, and conference papers, effectively communicating research insights to multidisciplinary audiences
  - Utilize Python, Matlab, C#, and Java for data engineering, modeling, and analytical tasks

- **RESEARCH INTERN| ECOLE D'INGÉNIEURS GÉNÉRALISTES, DE L'INFORMATIQUE À L'ÉLECTRONIQUE (ESEO), FRANCE| JUNE 2023 – DECEMBER 2023**
  - Implemented big data frameworks: Successfully deployed Apache Hadoop, Apache Hive, and Apache Spark to establish a big data processing environment
  - Database Management: Created and managed a TPC-H database for benchmarking OLAP processes, integrating Apache Hive for structured data storage and manipulation
  - OLAP Data Cube Deployment: Deployed and managed an OLAP data cube in Hive, enabling advanced data analysis capabilities
  - Machine Learning Algorithms Deployment: Applied machine learning algorithms over big OLAP data cubes

- **SIGNAL PROCESSING AND RECORDING SPECIALIST | NATIONAL BRAIN MAPPING LAB (NBML), IRAN | APRIL 2019 – JANUARY 2022**
  - Led EEG signal processing unit; designed protocols and built analysis pipelines (EEGLAB/MNE).
  - Conducted ERP/time–frequency/connectivity analyses and trained ML classifiers for neuro/behavioral prediction
  - Designed experimental tasks using Psychtoolbox and PsychoPy
  - Managed EEG data recordings for various applications including EEG-VR, EEG-fNIRS, EEG-TMS, and etc

## Education

- **PH.D. IN INFORMATION AND COMMUNICATION TECHNOLOGY (ICT) | MARCH 2025| UNIVERSITY OF CALABRIA, ITALY**
  **Dissertation:** *Multidimensional Big Data Analytics for Knowledge Discovering and Analysis from Big Web Knowledge Bases*
  **Supervisor:** Prof. Alfredo Cuzzocrea
- **M.S. IN BIOMEDICAL ENGINEERING | JULY 2017 | AMIRKABIR UNIVERSITY OF TECHNOLOGY, IRAN**
  **Dissertation:** *Quantification of Depression Disorder Using EEG Signal*
  **Supervisor:** Prof. Mohammad Hassan Moradi
- **B.S. IN ELECTRONICS ENGINEERING | SEPTEMBER 2014 | BABOL NOSHIRVANI UNIVERSITY OF TECHNOLOGY, IRAN**
  **Dissertation:** *Design and Implementation of A Digital Compass*
  **Supervisor:** Prof. Mehdi Ezoji

## Skills

- **Machine Learning & AI**
  - scikit-learn, PyTorch, TensorFlow/Keras
  - classical ML, transformers, Graph Neural Networks, reinforcement learning
  - Explainable AI (SHAP, feature importance)
  - LLMs (LLaMA, LangChain, RAG)
- **Data/Systems:**
  - Spark, Hive, Hadoop
  - AWS, GCP
- **Dev/Research Workflow:** Git, Linux, Docker

- **Programming:** Python, Matlab, Java, C#
- **Biomedical Signals & Medical Imaging**
    - PyRadiomics, SimpleITK
    - Signal filtering methods & feature engineering

## Publications

- [In Progress] Mojtaba Hajian, Somayeh Khosravi Khorashad, Sepideh Khoneiveh, Reza Khosrowabadi, "**Decoding Neural Dynamics of Suspense: EEG Connectivity and Power Analysis with Machine Learning for Pre- and Post-Suspense State Classification**", *Applied Soft Computing*, 2026.
- Mojtaba Hajian, Mohammad Hassan Moradi, '**Quantification of Depression Disorder Using EEG Signal**', *The 24th National and the 2nd International Iranian Conference on Biomedical Engineering (ICBME 2017)*, Tehran, Iran, 2017.
- Yousef Mohammadi, Mojtaba Hajian, Mohammad Hassan Moradi, '**Discrimination of Depression Levels Using Machine Learning Methods on EEG Signals**', *The 27th Iranian Conference on Electrical Engineering (ICEE 2019)*, Yazd, Iran, 2019.
- [Submitted] Alfredo Cuzzocrea, Mojtaba Hajian, "**Multidimensional Clustering over Big Data: Models, Issues, Analysis, Emerging Trends**", *Journal of Big Data*, Springer, 2026.
- [In Progress] Alfredo Cuzzocrea, Mojtaba Hajian, "**Data Cube Compression at Work! Effective and Efficient OLAP over Big Datasets in Mobile Cloud Environments**", *Information Systems*, 2026.
- Alfredo Cuzzocrea, Mojtaba Hajian, "**ClassCube: Effective and Efficient Big OLAP Data Cube Classification via Dimensionality Reduction**", *ICEIS*, 2024.
- Alfredo Cuzzocrea, Mojtaba Hajian, Abderraouf Hafsaoui, **"MALAGA – MultidimensionAL Big DAta Analytics over Massive Graph DAta"**, *IEEE International Conference on Big Data* (2024), Washington DC, USA, 2024.
- Alfredo Cuzzocrea, Mojtaba Hajian, "**Experimental Analysis and Assessment of a Real-Life Cloud Mobile Big OLAP System Enhanced with Compression and Approximation Paradigms**", *IEEE International Conference on Big Data* (2024), Washington DC, USA, 2024.
- Alfredo Cuzzocrea, Mojtaba Hajian, Abderraouf Hafsaoui, "**Compressing Big OLAP Data Cubes over Mobile Clouds: A Hierarchy-Based Data Partitioning Approach**", *Proceedings of the 32nd Symposium of Advanced Database Systems, Villasimius, Italy,*, 2024.
- Somayeh Khosravi Khorashad, Sepideh Khoneiveh, Mojtaba Hajian, Reza Khosrowabadi, '**Decoupage Parameters Related to the Suspense Points and Symmetrical Activation of Brain Waves at the Frontal Lobe: An EEG Study of the Movie 'Psycho'** ', *Quarterly Review of Film and Video*, 2022.
- A Cuzzocrea, M Hajian, '**Effective and Efficient Big OLAP Data Cube Compression in Mobile Cloud Environments: The IQTS Algorithm**', *IEEE International Conference on Big Data (2023)*, Sorrento, Italy.
- A Cuzzocrea, P Figuera, M Hajian, PG Bringas, '**A Theoretical Framework for Supporting Clustering Validation via Non-Negative-Matrix-Factorization Trace Sequences Over Probabilistic Spaces**', *IEEE Conference on Cloud and Big Data Computing (CBDCom 2023)*, Abu Dhabi, United Arab Emirates.
- Alfredo Cuzzocrea, Carson Leung, Mojtaba Hajian and Marshall Jackson, '**Effectively and Efficiently Supporting Predictive Big Data Analytics over Open Big Data in the Transportation Sector: A Bayesian Network Framework**', *The 8th IEEE International Conference on Cloud and Big Data Computing (CBDCom 2022)*, Calabria, Italy.

- Alfredo Cuzzocrea, Christophe Cerisara, Mojtaba Hajian, '**Risk Analysis for Unsupervised Privacy-Preserving Tools**', *Proceedings of the 30th Italian Symposium on Advanced Database Systems, SEBD 2022, Tirrenia (Pisa)*, Italy, June 19-22, 2022.

## Research Interests

- Machine learning and deep learning
- Representation learning and feature abstraction
- Generative AI (LLMs, VLMs, Diffusion Models)
- Supervised and weakly supervised learning
- Explainable and interpretable artificial intelligence (XAI)

## Teaching Experiences

- "Knowledge Transfer Activity at Paracelsus Medical University (Salzburg)", Workshop, Provided methodological training and support to medical physics experts, Salzburg, Austria, September 2025. Topics Covered:
    - Image preparation and automated lesion segmentation workflow
    - Radiomics feature extraction
    - Feature organization and dataset construction
    - Machine learning modeling strategies for small datasets
- "Machine Learning Applications for Biomedical Signal Processing", Webinar, Brain Mapping Laboratory, Tehran, Iran, February 2021.
- "The 16th workshop on EEG Signal Recording, Processing and Analyzing", Workshop, National Brain Mapping Laboratory, Tehran, Iran, October 2020.
- "Instructor of over 50 trainees in the EEG data analysis and recording internship course", Tutor, National Brain Mapping Laboratory, Tehran, Iran, August 2019-December 2021.
- "An introduction to the function of the EEG amplifier and how to use it", Seminar, Brain Mapping Laboratory, Tehran, Iran, January 2020.
- "Robotics and micro-controllers" Workshop, Babol Noshirvani University of Technology, Mazandaran, Iran, July 2010-September 2013.

## Honors and Awards

- Commendation for Establishing EEG Setup for Sleep Services
    - National Brain Mapping Laboratory (NBML) | 2020
- Team Leader Award for Manufacturing Electronic MRI-Compatible Response Grips
    - National Brain Mapping Laboratory | 2020
- Reviewer for 'Frontiers in Biomedical Technologies' Journal and 'IEEE Big Data' Conference
- 2nd Place in Smart City Startup Weekend (MOPEC-2017)
    - Sponsored by Municipality of 5th District of Tehran City | 2017
    - References**Available upon request.**